

Summary

AI-focused software architect with over a decade of full-stack engineering experience, now specializing in agentic systems, LLM-powered developer tooling, and spec-driven AI development pipelines that have measurably accelerated delivery by up to 4x. Adept in architecture design, overseeing implementations, mentoring teams, designing multi-agent AI orchestration, maintaining strong client relationships, and acquiring deep domain expertise. Committed to translating emerging AI capabilities into reliable, production-grade systems that expand what one engineer can deliver.

Languages: Java (6 through 24), Lua, C#, C++, JavaScript, Python, Bash, PowerShell, Lisp

Frameworks: Spring, Spring Boot, Spring Data, JUnit, Mockito, ReactJS, NodeJS, jQuery

Databases: MySQL, PostgreSQL, MongoDB, MSSQL, Oracle, DB2, Hibernate

Design: Microservices, Test Driven Development, REST, Architecture

Others: Docker, Tomcat, Caddy, Linux Administration, Gradle, Maven, Ant, OAuth2, OpenShift, Kafka, IBM MQ

AI Tooling: GitHub Copilot, Claude Code, Claude Agent SDK (Python & TypeScript), Cursor, Prompt Engineering, Agentic Orchestration & Subagent Design, Spec-Driven Development, Skill Development, Hook Development, and MCP Server Development

Experience

AI Architect

InRhythm, O'Fallon, IL

January 2026 – Present

Environment: Java (21), Spring Boot, Spring Batch, Amazon EKS, GitHub Copilot

- Designed and deployed a Spec Driven Development AI DLC pipeline to modernize legacy pSQL Oracle batch jobs running on the database host box to individual containerized Spring Batch jobs running in EKS. The pipeline sped up development by ~4x (one quarter per batch to 1 month per batch).
- Generalized the above pipeline to modernize a Struts-based JSP legacy monolithic webapp into a modern Angular front end backed by domain specific Spring Boot microservices. This improved development times by ~2x.
- Developed a reflexion memory system for the pipeline for it to learn from its mistakes and learn the client's patterns and architecture better.

Senior Software Engineer

Bank of America, Chicago, IL

April 2024 – April 2025

Environment: Java (17), Spring, Spring Boot, DB2, OpenShift, Kubernetes, Git, TDD, Kafka

- Contributing senior developer on a SAFE agile team of 6 developers, 3 onshore and 3 offshore (in India).
- Added new REST endpoints to a microservice for manipulating ACH transfer limits, vastly speeding up changes to them by eliminating the need for an entire application rebuild.
- Added resiliency in ACH fraud detection by adding circuit breakers to save resources and using the store and forward pattern with Kafka to store the transactions that were unable to process during downtime.

Senior Software Engineer

PNC Bank, Chicago, IL

March 2023 – April 2024

Environment: Java (11), Spring Boot, Spring Data, MySQL, Docker, Git, TDD

- Led a team of three building a Java 21 Spring Boot microservice with Kafka and MySQL to bridge a new REST-based payment gateway with legacy batch applications.
- Applied TDD and modern design patterns for a performant, extensible, fault-tolerant, and maintainable product.
- Ensured PNC DevOps compliance, including CI/CD on internal RHEL OpenShift and full monitoring/logging coverage.

Senior Software Engineer

Loop Commerce, Columbia, MO

October 2022 – December 2022

- Loop Commerce was shut down by its parent company, Synchrony Bank, one month after I was hired.

Senior Software Engineer

SuprTek, O'Fallon, IL

May 2018 – August 2022

Environment: Java (11), Spring, JavaScript, Node.js, React, MongoDB, Git, TDD

- Led a team of 8 converting a legacy, spreadsheet-driven logistics line of business as part of a full-scale USTRANSCOM application rebuild.
- Designed the data, service, REST API, and messaging layers for the Coronet line of business using MongoDB, Java 17, Spring Boot, and Kafka.
- Served as interim Product Owner for six months, leading client interactions and advocating for the client in team decisions.

Senior Software Engineer

DMI, Carmel, IN

November 2017 – March 2018

Environment: Docker Enterprise, Red Hat Linux, Bash, JSON, Artifactory, Jenkins, Microsoft Team Foundation Server, Maven, SED

- Managed the transition of Allison Transmission's DevOps infrastructure from external DMI hosting to internal hosting.
- Used Docker to migrate Jenkins, Team Foundation Server, Artifactory, and Git servers.
- Built a Docker image and Maven repositories, migrating existing artifacts into the new Artifactory server.

Healthcare Informaticist

BJC Healthcare, St. Louis, MO

November 2014 – July 2017

Environment: Java (6, 7, & 8), Spring, Struts, JavaScript, jQuery, Sybase, DB2, SQL Server, PostgreSQL, Python, Linux, Bash, Git, CVS, Tomcat, MQ Series, Maven, Ant, Jira (administration and use)

- Led design and implementation of BJC's CDC grant-funded software solution for antibiotic use and resistance reporting.
- Led conversion of major projects' build infrastructure to Maven, mentoring 6 team members on its use.
- Maintained and updated hospital-acquired disease reporting infrastructure using Java (6, 7, & 8), Spring, Struts, JavaScript, and jQuery.
- Interpreted CDC technical guides to update and implement new XML reports for CDC submission.

AI Projects & Ventures

Margo — Agentic Chief-of-Staff Assistant

Ongoing

Environment: Claude Agent SDK, Python, Cloudflare Workers, Twilio, Notion

- Architected a persistent AI orchestrator that loads purpose-built subagents just-in-time and runs proactive, local-first loops to screen calls, texts, and email and manage daily scheduling.
- Designed an "act-low / ask-high" approval gate so the system autonomously handles safe actions while drafting and holding anything outbound for review.

Spec-Driven AI Development Pipeline

Personal R&D

Environment: TypeScript, zod, SQLite, Ollama, Claude, Git Flow

- Built a self-verifying, spec-first engineering pipeline with a durable task-graph runtime and typed contracts, using a dedicated verification gate so the agent doing the work never grades its own output.
- Authored a reusable suite of 7 Claude Code skills, including context-loaders and crash recovery, to make multi-agent engineering repeatable.

AI-Native CRM & Call Screener

Personal Venture

Environment: Java 21, Spring Boot 3, React, TypeScript, Cloudflare Workers, Twilio, Claude, ElevenLabs

- Designed and built solo a multi-tenant customer-engagement platform with an AI intake pipeline that transcribes voicemails into structured leads, triages submitted photos, and drafts review replies.
- Built a real-time telephony screener wired to an LLM persona that determines caller intent and routes only what needs direct attention.

Education

South Dakota School of Mines and Technology, Rapid City, SD
B.S. in Computer Science

August 2010 – December 2014